KingdomForge: Benchmark Hardening & Live-Run Protocol

Hardened Reinforcement Learning & Agentic Decision-Making Benchmark for LLMs

Presenter Team Submission Deck | 10-Minute Evaluator Review

Core Problem	Benchmark v2 Scope	Live-Run Integrity
Evaluating LLM planning reliability, resource budgeting, and recovery under strict live-run time & budget constraints.	13 capability-centric tasks: Tasks 1–6 (Frozen v1 regression) + Tasks 7–13 (Distinct planning & optimization dimensions).	Enforces monotonic 16-minute slot budget, 10-tool/reply cap, 60s tool timeouts, discrepancy detection, & isolated verifier.

Key Validation Status: 112/112 Unit Tests Passed | 26/26 Baseline Gates Passed (100%) | Monotonic Live Caps Verified

The Challenge: LLMs in Interactive Constrained Environments

Why Typical Benchmarks Fail to Catch Critical Planning Breakdowns

- **Illusion of Competence:** Models often self-report success or execute redundant tool queries without altering environment state.
- **Resource Hallucination & Deadlocks:** Inability to forecast opportunity costs leads to early depletion of non-renewable resources (e.g. gold).
- **Error Corruptibility:** Rejected actions frequently induce repetitive failure loops rather than adaptive recovery.
- **Evaluator Flakiness:** Flaky baselines, state leakage across resets, or unverified agent claims compromise benchmark authority.

KingdomForge's Guiding Principles:

1. **Zero Trust in Agent Claims:** Verification is 100% programmatic against actual environment state.
2. **Monotonic Clock Hard Caps:** Strict enforcement of 16-min slot, 10-min episode, 2-min baseline, and 60s tool call limits.
3. **Discrepancy Transparency:** Immediate detection and classification of state-mutation errors and invariant breaches.

System Architecture & Live-Run Protocol Execution

Isolated Execution Pipeline with Two-Run Protocol Compliance

Stage 1: Preflight & Gate	Stage 2: Model Episode	Stage 3: Verifier & Scoring
• Load configuration & APPEND_FILE (exact verbatim append) • Run Reference Baseline (≥ 0.95) • Run No-op Baseline (≤ 0.05) • Combined timeout: ≤ 120s • If gate fails, model is NOT called	• Model loop: Max 40 replies • Reply budget: Max 10 tools/reply • Snapshot tool isolation (60s timeout) • State invariants checked after each turn • Running reward recomputed on each step	• Programmatic isolated verifier • Max scoring time: ≤ 120s • Failure classification strictly partitioned • Invariants violation overrides score to 0.0 • Structured JSONL telemetry written

Two-Run Evaluation Protocol: Run A executes the README demo task (task_3). Run B supports judge-selected tasks with verbatim appended instructions from APPEND_FILE. Both use identical harness integrity gates.

Benchmark v2: 13 Distinct Capability Dimensions

Comprehensive Multi-Axis Evaluation Matrix

Tier	Task ID & Name	Target Capability Dimension	Key Constraint / Rubric
Frozen v1	Task 1: Basic Farming	Basic Resource Building	Build farms, stockpile food
Frozen v1	Task 2: Wood Economy	Worker Allocation Dynamics	Assign workers to lumber mills
Frozen v1	Task 3: Market Trading	Commercial Exchange & Scaling	Build market, trade surplus for gold
Frozen v1	Task 4: Fortified Settlement	Defensive Preparation	Walls & towers to repel raids
Frozen v1	Task 5: Prosperous Realm	Multi-Tier Population Expansion	Expand housing & maintain nutrition
Frozen v1	Task 6: Master Architect	Advanced Multi-Sector Realm	Balanced multi-structure optimization
v2 New	Task 7: The Mason's Citadel	Dependency & Action Ordering	Workshop → Masonry → Wall → Castle
v2 New	Task 8: Scarce-Resource Budgeting	Austerity & Opportunity Cost	Zero starting gold; prioritize timber export
v2 New	Task 9: The Long Horizon	Multi-Turn Temporal Planning	Survive 15+ turns without starvation cliff
v2 New	Task 10: Capacity Expansion	Workforce & Housing Trade-offs	Expand housing capacity to 12+, pop to 9+
v2 New	Task 11: Adaptive Recovery	Fault Recovery & Replanning	Diagnose 0 stone, mine quarry, rebuild walls
v2 New	Task 12: Crown of the Realm	Integrated Constrained Planning	Simultaneous masonry, defense, & castle
v2 New	Task 13: Economic Optimization	Optimality Awareness & Limits	Feasibility verification rubric & stopping

State Invariants & Explainable Reward Breakdown

Deterministic Accounting with Explicit Loss Explanations

State Invariants (Validated after every state mutation):

- **Non-Negative Resources:** gold ≥ 0, food ≥ 0, wood ≥ 0, stone ≥ 0. Resources cannot underflow.
- **Population Conservation:** total_population == idle_workers + sum(assigned_workers). Workers cannot vanish.
- **Worker Capacity Caps:** assigned_workers[b] ≤ count[b] * cap[b]. Assignments cannot exceed building facilities.
- **Non-Negative Defense:** defense ≥ 0. Raids degrade fortifications but never induce negative values.

Explainable Reward Decomposition:

Every step and episode logs a structured dictionary containing:

- **Objective Milestones:** Discrete boolean/proportional breakdown per task requirement.
- **Invariant Audit:** Any violation triggers immediate integrity_violation and sets score to 0.0.
- **Discrepancy Flags:** Captures pre-state hash, post-state hash, and action arguments.

Independent Verifier & Discrepancy Detection Engine

Automated Anomaly Detection & Safe Recovery Policies

Discrepancy Category	Trigger Condition	Automated Recovery / Enforcement
Error with State Mutation	Tool returns 'Error:' but state hash changes.	High severity flag logged; rollback snapshot prevents corruption.
State Invariant Violation	Physical laws violated (e.g. negative gold).	Critical severity; verifier forces episode score to 0.0.
Provisional Mismatch	Step score contradicts recomputed verifier score.	Final verifier overrides provisional claims unconditionally.
Rogue Tool Stalling	Single tool call exceeds 60s timeout.	Worker thread abandoned; snapshot abandoned; state unharmed.
Context Window Overflow	History exceeds 60,000 tokens.	Conversation history pruned while strictly preserving tool-call pairs.

Model Feedback: Rejected actions return structured guidance: INVALID_ACTION: <tool> | Reason: <cause> | Constraint: <limit> | State change: none | Recovery: <actionable advice>.

Live-Run Enforcement & Concurrency Controls

Monotonic Clock Enforcement of Evaluator Hard Limits

All execution timers utilize Python's monotonic clock (time.monotonic()), completely immune to system clock adjustments or NTP skew.

Component / Slot Phase	Hard Budget Limit	Observed KingdomForge Baseline Execution
Clean Docker Image Build	3 minutes (180s)	< 45 seconds (lean debian-slim with cached python wheels)
Combined Baseline Gate	2 minutes (120s)	**0.08 seconds** combined across all 13 tasks (1,500x faster than limit)
Model Episode Duration	10 minutes (600s)	Real model episodes complete in 120–240 seconds
Max Replies per Episode	40 replies	Strictly enforced; terminates with termination_reason='step_cap'
Max Tool Calls per Reply	10 calls max	Calls ≥ 10 rejected with explicit diagnostic feedback
Per-Tool-Call Timeout	60 seconds	Non-blocking ThreadPoolExecutor; state snapshot abandoned on timeout
Overall Live Slot Cap	16 minutes (960s)	Global slot hard stop enforced across baselines, model, and verifier

Security Architecture & Threat Model Defences

Defensive Engineering Against Rogue Actions and Accidental Disclosure

- **Zero Shell/Code Execution:** Model interacts exclusively through 9 typed simulation tools. No arbitrary code execution, filesystem browsing, or command injection primitives exist.
- **Secrets Redaction:** Logger sanitizes all API keys (sk-..., bearer tokens, passwords) before writing to JSONL or stdout.
- **Safe APPEND_FILE Handling:** Non-existent or empty APPEND_FILE halts execution immediately with EXIT_CONFIG_ERROR (3). Judge instructions are appended verbatim without reinterpretation.
- **State Isolation Across Episodes:** Every episode instantiates an isolated KingdomEnv instance with clean state and reproducible PRNG seed.
- **Timed-Out Thread Quarantine:** When a tool exceeds 60 seconds, the worker thread is terminated in non-blocking mode; mutated snapshot state is garbage collected without affecting the live environment.

Package Audit: Submission archive strictly excludes .env, API keys, private tokens, virtual environment binaries, and build artifacts.

Empirical Results & Real Evaluation Evidence

Measured Performance Across Benchmark Suite & Luna Model Runs

Automated Test Suite Results:

- **112 Tests Executed:** 112 Passed, 0 Failed, 0 Regressions. Execution time: 1.09s.
- **Baseline Gate Checks (gate.json):** 26/26 gates passed (13 Reference @ 1.0000, 13 No-op @ 0.0000).

Authentic Model Trajectories Captured (sample-trajectories/):

- **Real Successful Trajectory (success_trajectory_task_3.json):** Model gpt-6-luna completed Task 3 in 27 steps. Successfully planned market construction, wood assignment, surplus trading, and worker recruitment. Final score: **1.0000** (PASSED).
- **Real Failure Trajectory (failure_trajectory_task_7.json):** Model gpt-6-luna on Task 7 (The Mason's Citadel). Successfully built workshop, market, and fortified walls, but miscalculated long-term stone budgeting for castle construction before the 40-step limit. Final score: **0.6500** (FAILED - authentically demonstrating capability boundary under strict prerequisites).

Submission Readiness & Compliance Checklist

Full Alignment with Hackathon Evaluation Protocol

Evaluation Requirement	KingdomForge Status	Evidence / Verification Path
Live-Run Protocol Limits	**FULLY ENFORCED**	Monotonic clock guards for 16-min slot, 10-min run, 40 replies, 10 tools/turn.
Two-Run Protocol (Run A & B)	**VERIFIED**	Run A: Task 3 demo. Run B: Judge-selected task with exact APPEND_FILE.
Benchmark v1 Frozen Suite	**PASSED**	Tasks 1–6 regression verified; 100% baseline pass rates preserved.
Benchmark v2 (Tasks 7–13)	**VALIDATED**	7 distinct capability dimensions implemented, baselined, & unit-tested.
Explainable Reward Breakdown	**IMPLEMENTED**	Decomposed reward logging; failure classification strictly partitioned.
Discrepancy Detection Engine	**ACTIVE**	Catches error-with-mutation and invariant violations instantly.
Structured Error Feedback	**COMPLIANT**	Model receives actionable recovery guidance on rejected actions.
Required Artifacts	**READY**	gate.json, test-run-logs/, sample-trajectories/, design-doc.pdf, deck.pdf.